Department of Environmental and Geographical Science
University of Cape Town

EGS3012S – ATMOSPHERIC SCIENCE

TEST 1: 15 August 2025
Time: 08:00 – 08:45
Total: 55 Marks
Internal Examiner: Dr Phikolomzi Matikinca

INSTRUCTIONS

- This test has **TWO** sections (A and B).
- Section A consists of Multiple-Choice Questions.
- Section B consists of Short Answer Questions.
- Answers all the questions.

SECTION A: MULTIPLE CHOICE QUESTIONS [20 Marks]

**Answer ALL questions in this section.*

1. A farming community notices that rainfall now arrives 3 months later than it did 30 years ago. Crops are failing due to mismatched flowering and rainfall. Which concept best explains this situation?
A. Climate oscillation
B. Biodiversity loss
C. Phenological shifts
D. ENSO neutral phase

2. What role do climate models play in understanding climate change?
A. They simulate climate responses under different greenhouse gas scenarios
B. They calculate daily weather forecasts
C. They monitor volcanic ash distribution
D. They reduce global carbon emissions

3. How does global warming affect ENSO events?
A. It reduces the frequency of all ENSO phases
B. It prevents La Niña conditions from occurring
C. It raises sea surface temperatures, intensifying El Niño events and altering global rainfall patterns
D. It cools the tropical Pacific Ocean, weakening the ENSO cycle

4. Which of the following is NOT correct about how climate sensitivity is estimated?
A. Examining past climate changes and their potential drivers
B. Simulating the climate system to predict how it will respond to different scenarios of GHG emissions
C. Analyzing the complex interactions within the climate system that amplify or dampen the initial warming
D. Measuring daily weather patterns to directly calculate long-term climate sensitivity

5. Which of the following marine ecosystems is highly vulnerable to acidification and supports 25% of marine life?
A. Coral reefs
B. Mangrove
C. Deep Sea vents
D. Kelp forests

6. A study along the South African west coast found that lower pH in winter significantly reduces availability of carbonate ions in seawater. Which of the following will be most significantly impacted?
A. shell-building organisms
B. coral reef photosynthesis
C. fish migration
D. ocean current

7. What is the main function of the Earth's atmosphere in relation to solar radiation?
A. It blocks all radiation from entering
B. It reflects all sunlight back into space
C. It traps radiation from the Earth's core
D. It shields life from harmful solar radiation

8. In marine benthic communities, reduced recruitment due to climate-induced ocean acidification is likely to lead to which of the following outcomes at the community level?
A. Expansion of ecological niches
B. Increased top predator abundance
C. Shift in community composition favoring stress-tolerant species
D. Stable population age structures

9. Ocean acidification is mainly driven by __________________.
A. excess nutrient runoff
B. oil spills
C. CO_2 absorption by seawater
D. oceanic carbon cycle

10. Which process amplifies initial warming by reducing reflectivity of Earth's surface?
A. Evapotranspiration
B. Carbonate buffering
C. Ice-albedo feedback
D. ENSO oscillation

SECTION B: SHORT ANSWER QUESTIONS [35 Marks]

Answer **ALL questions in this section.*

QUESTION 1

Examine the **Figure 1** below and answer the questions that follow.

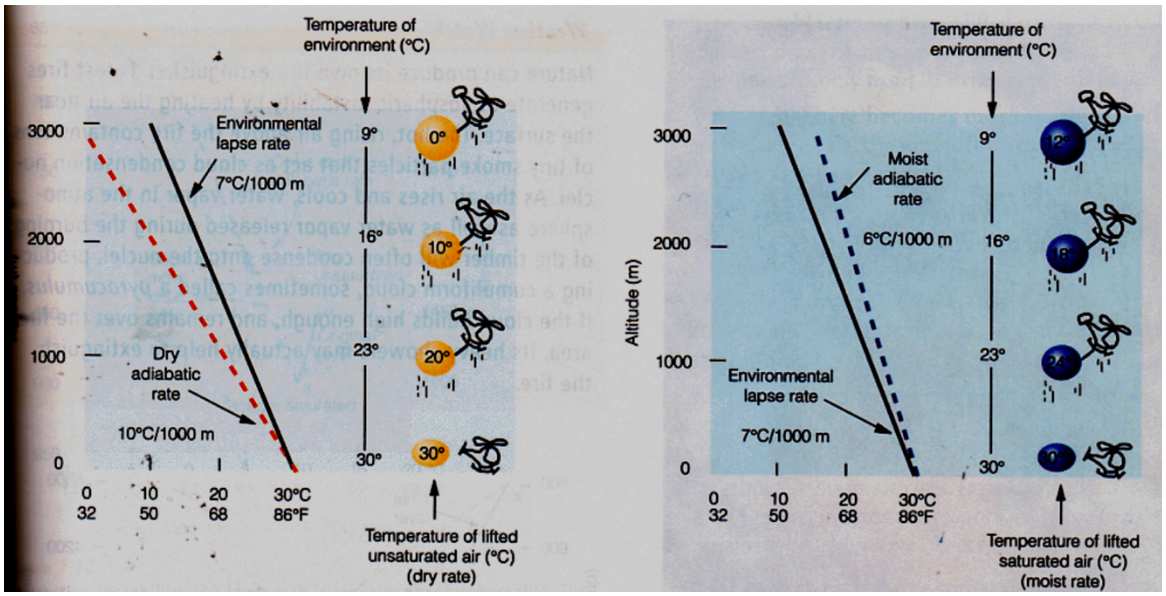

Figure 1: Atmospheric condition

1. What atmospheric condition is depicted in this figure? **[2 Marks]**. Explain your answer. **[2 Marks]**

    Conditionally unstable atmosphere *(2 Marks)*

    This is because the ELR (rate of temperature decrease of the surrounding air) is greater than the MALR but less than the DALR *(2 Marks).*

    Some students might write it like this: MALR < ELR < DALR. Award the marks.

2. Based on this atmospheric condition, describe what happens as the unsaturated and saturated air parcels rise in relation to the surrounding air. **[8 Marks]**

    An unsaturated air parcel cools at the DALR and remains cooler than the surrounding air, so it sinks back *(2 Marks)*. Relatively cool air is heavier and denser. This suppresses vertical motion of rising air, causing it to sink back *(2 Marks*).

    A saturated air parcel cools more slowly at the MALR because the condensation of water vapor releases latent heat, which partially offsets the cooling, and could become warmer and less dense than the surrounding environment, continuing to rise *(2 Marks*). Relatively warm air is lighter, and this promotes vertical motion of rising air (*2 Marks)*.

3. What weather phenomena is associated with this atmospheric condition? **[2 Marks]**

    Cumulus cloud formation, localized thunderstorms, and convective showers can all occur when the right trigger lifts air parcels to saturation.

QUESTION 2

Climate predictions and climate projections are both used to understand future climate conditions but differ in scope and purpose. Explain this difference. **[4 Marks]**

Climate predictions

Short-term forecasts (months to a few years) of specific climate phenomena, like seasonal temperature anomalies or precipitation patterns, based on current climate conditions and natural variability. *(2 Marks)*

Climate projections

Long-term simulations (decades to centuries) of how the climate might change under different scenarios of greenhouse gas emissions and socioeconomic development factors, exploring potential future climates rather than predicting specific events. *(2 Marks)*

QUESTION 3

Figure 2 below, taken from a study done by Woods et al. (2022), is a Thermal Performance Curve (TPC) showing that the feeding rates of leaf-mining caterpillar larvae depend on temperature. The study found that the feeding rates were highest at 25°C. Grey dots and grey lines are for individual larvae, and **the black line is the fitted thermal performance curve** (N = 39 larvae). The dot and error bars at 40°C estimate the mean and 95% confidence interval. Examine **Figure 2** and answer the questions that follow.

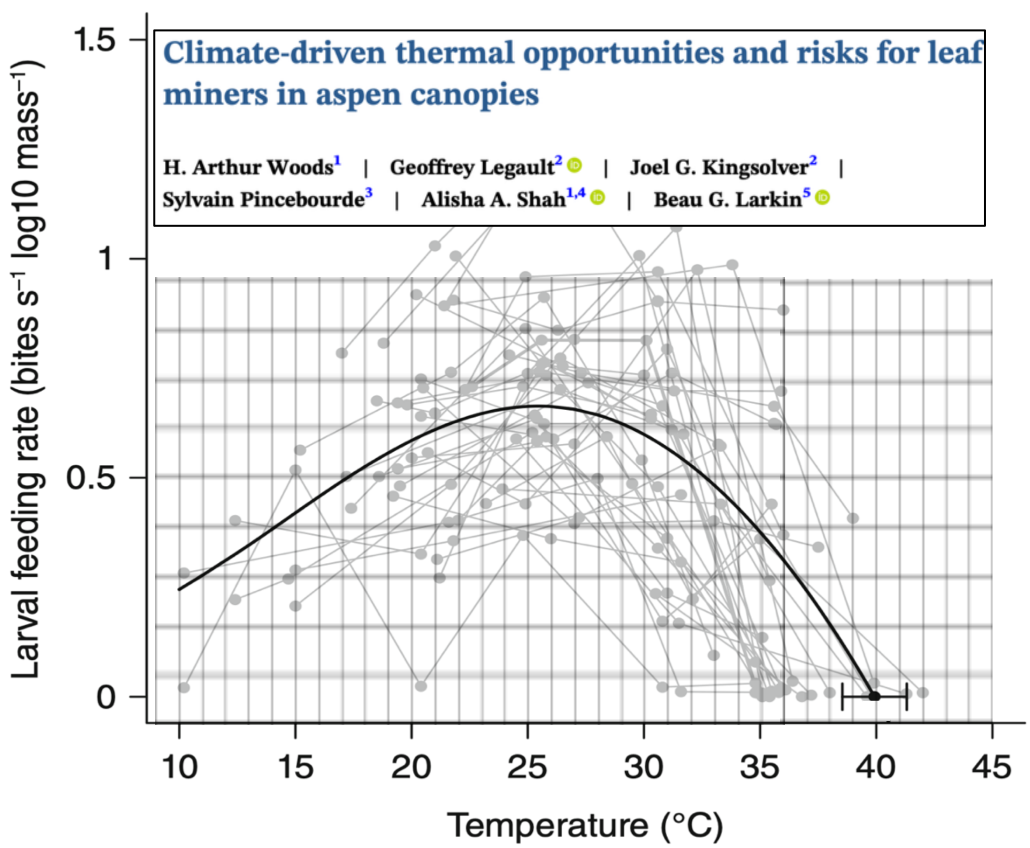

Figure 2: Thermal Performance Curve (TPC) for temperature dependence of feeding rates by the leaf-mining caterpillar larvae. The grid has been added for scale purposes only.

1. Calculate the following and include the formulas and units where applicable:

a) Thermal tolerance range. **[3 Marks]**

CTmax – CTmin = 40°C - 10°C = 30°C *(1 mark for each step). Deduct 0.5 if there are no units.*

b) Thermal Safety Margin. **[3 Marks]**

CTmax – Topt = 40°C - 25°C = 15°C *(1 mark for each step). Deduct 0.5 if there are no units.*

2. Consider a current (ambient) temperature of 28°C and a 3°C projected climate warming scenario. Calculate the following and include the formulas and units:

a) Relative Sensitivity to Warming (RSW) at current temperature. **[3 Marks]**

(1 mark for each step). Deduct 0.5 if there are no units.

(Tcurrent – Topt) ÷ (CTmax – CTmin)

= (28°C – 25°C) ÷ (40°C – 10°C)

= 0.1°C

b) RSW considering the projected warming scenario. **[3 Marks]**

(1 mark for each step). Deduct 0.5 if there are no units.

(Twarming – Topt) ÷ (CTmax – CTmin)

= (31°C – 25°C) ÷ (40°C – 10°C)

= 0.2°C

c) Based on the calculated RSWs, will the feeding rates be affected by climate warming? **[1 Marks]** Explain your answer. **[4 Marks]**

Yes – the feeding rates will be affected by climate warming *(1 Mark)*

This is because the Relative Sensitivity to Warming calculated for the warming temperature is greater/higher/bigger than the RSW calculated for the current/ambient temperature *(2 Marks)*. Therefore, the feeding rates will be more sensitive to climate warming in the future *(2 Marks)*.